A Ceiling Dark Matter Cannot Impose: the Bounded-Boost Theorem for MOND-Class Kernels, and What It Says About Galaxies and Clusters

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Abstract

Any modified-gravity kernel that writes the observed acceleration as $g = \nu(y)\, g_N$, $y = g_N/a_0$, with a Newtonian limit $\nu \to 1$ and a deep-MOND limit $\nu \to y^{-1/2}$, makes a prediction that a dark-matter halo structurally cannot make: the *acceleration excess* $\Delta \equiv g_{\rm obs} - g_{\rm bar}$ is bounded above by a pure number times a_0. For the exponential carrier $g_{\rm bar} = g\,(1 - e^{-g/a_0})$ the bound is exact and elementary, $\Delta \le a_0/e = 0.367879\ldots a_0$, attained at $g_{\rm bar} = (1 - 1/e)a_0$; the completed kernel saturates there, so the bound is attained on a plateau. Across five standard kernels the constant lies in $[0.250, 1.000]$, so the ceiling is kernel-independent up to an $O(1)$ factor. In a halo model the same quantity is $g_{\rm halo}$, set by M_{200} and concentration, which span decades and are not tied to a_0; the ceiling is therefore a one-sided hard wall with no fitting freedom, and a single system above it falsifies the kernel. We test it. (i) On 2352 SPARC rotation-curve points beyond 2 kpc from 144 galaxies (the survey's own $Q < 3$, $i > 30°$ cuts), 99.23% obey the widest kernel bound at both a_0 footings; the eighteen $> 3\sigma$ exceptions lie in five named galaxies, thirteen of them in NGC 5985 alone. (ii) The ceiling is tight enough to discriminate between kernels: 9.8% of those points (263, spread over 32 of 144 galaxies) exceed the exponential carrier's own ceiling a_0/e at $> 3\sigma$, and between $g_{\rm bar} = 1$ and $4.5\,a_0$ the measured excess is 2–3× that ceiling against 1.2–1.9× the wider $\nu_{\rm RAR}$ ceiling. The coherent stellar-normalisation freedom is independently pinned to +1.8% by the Newtonian limit, and raising a_0 cannot absorb the discrepancy (it would need 12× the canonical value). (iii) In X-COP cluster cores, after correcting a radius-unit error we verify directly from the FITS headers, the ceiling fails by a factor 9.1 against the exponential carrier's bound and 3.4 against the widest bound of any kernel, so no choice of interpolation function can absorb it. Both escapes close quantitatively: nonthermal support would need $\sigma_{\rm 1D} = 673$–$1155\,{\rm km\,s^{-1}}$ against the Hitomi/XRISM Perseus measurement of 164 ± 10, and unseen baryons would need 3.9–5.2× the directly imaged X-ray gas. Clusters therefore require a genuine extra source *as a theorem rather than a fit.* The required source is baryon-tracing, but so is the published NFW fit, so its profile shape does not discriminate; we report that as a null. This is a preprint of an AI-assisted draft. It claims no completed relativistic theory, no derivation of a_0 or of $\kappa = \frac{1}{2}$, and no empirical detection.

1 The action and the kernel

The candidate under test is general relativity plus a clock scalar τ with unit normal $n_\mu = -\partial_\mu\tau/\sqrt{-(\partial\tau)^2}$ and projector $q_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$, and one MOND scalar ϕ with $Q = n^\mu\partial_\mu\phi$, $V_\mu = q_\mu{}^\nu\partial_\nu\phi$, $Y = V\!\cdot\!V$:

$$S = \int d^4x\sqrt{-g}\Big\{\tfrac{1}{16\pi G}\,[R - 2\Lambda] - c_1(\nabla_\mu n_\nu)(\nabla^\mu n^\nu) - c_2(\nabla\!\cdot\! n)^2 - c_3(\nabla_\mu n_\nu)(\nabla^\nu n^\mu) + c_4(n\!\cdot\!\nabla n)^2 \quad (1)$$

$$+ 2(2-K_B)\, J^\mu \partial_\mu \phi - (2-K_B)\, \mathcal{J}\Big(Y + \xi^2 q^{\lambda\sigma} q^{\mu\nu} \nabla_\lambda V_\mu \nabla_\sigma V_\nu\Big) - K(Q)\Big\} + S_m[g, \psi_m],$$

with $J^\mu = n^\nu \nabla_\nu n^\mu$ the clock's four-acceleration, $K(Q) = K_2(Q-Q_0)^2$, $c_1 = -c_3 = K_B$ so that $c_{13} = 0$ and $c_T = c$, and matter minimally coupled. Nothing below depends on this particular action: the theorem applies to any kernel of the stated class. We carry both acceleration footings, $a_0 = 9.3619 \times 10^{-11}$ and $1.1279 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, throughout; $\kappa = \frac{1}{2}$ in $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ is fitted, not derived.

In the static weak-field limit the scalar carries an exponential kernel. Writing $y = g/a_0$ for the total field, the matter-sourced carrier obeys $g_{\rm bar} = g\,(1-e^{-y})$ for $y \le 1$, and above $y = 1$ the scalar force saturates at a_0/e.

2 The theorem

Define the *acceleration excess* in units of a_0,

$$\Delta(g_{\rm bar}) \;\equiv\; \frac{g_{\rm obs} - g_{\rm bar}}{a_0} \;=\; y\big[\nu(y) - 1\big]. \tag{2}$$

For the exponential carrier this is exactly

$$\boxed{\Delta = y\,e^{-y}, \qquad \frac{d\Delta}{dy} = (1-y)e^{-y}, \qquad \Delta_{\max} = \frac{1}{e} = 0.367879\ldots \text{ at } y = 1,} \tag{3}$$

verified symbolically: the critical point is unique, the second derivative there is $-e^{-1} < 0$, and both endpoint limits vanish. The maximum is reached at $g_{\rm bar} = (1-1/e)\,a_0 = 0.6321\,a_0$. Because the completed kernel saturates at $y = 1$, the excess does not fall back but stays at a_0/e: the bound is attained on a plateau.

The statement is not special to this kernel. Δ vanishes as $\sqrt{y}$ at small y and tends to 0 or a finite constant at large y whenever $\nu \to 1$ at least as fast as $1/y$; being continuous in between, it is bounded. Table 1 gives the supremum for five standard kernels, computed on a 2×10^6-point logarithmic grid.

kernel	$C = \sup\Delta$	at $g_{\rm bar}/a_0$
deep-MOND $\sqrt{\ }$, $g = \sqrt{a_0 g_N}$	0.2500	0.250
standard μ	0.3003	0.486
exponential carrier (this action)	0.3679	≥ 0.632 (plateau)
$\nu_{\rm RAR}$ (McGaugh et al. 2016)	0.6476	2.54
simple μ	1.0000	$y \to \infty$

Table 1: The ceiling is kernel-independent up to an $O(1)$ constant.

3 Why a dark-matter halo cannot make this prediction

In a halo model the same observable is $g_{\rm obs} - g_{\rm bar} = g_{\rm halo}$, the halo's own acceleration at that radius. It is set by the halo mass and concentration, which vary by orders of magnitude across the galaxy population and have no relation to a_0. There is no mechanism placing a common upper bound on $g_{\rm halo}$, still less one at a fixed multiple of an acceleration defined by the cosmological constant. A ceiling holding across every system and every radius is therefore a coincidence a halo model must absorb as a fine-tuning of the mass–concentration relation against the baryon distribution.

Two properties make this sharper than the radial-acceleration relation from which it descends. It is *one-sided*: only excursions above the wall count, so scatter, distance errors and inclination errors cannot manufacture a false confirmation, only a false violation. And it is *parameter-free* once a_0 is fixed: there is no amplitude, slope or scatter to fit, so a single system above the wall falsifies the kernel outright.

4 Test on rotation curves

We use the SPARC Rotmod curves with the survey's published sample cuts ($Q < 3$, $i > 30°$) and its fixed 3.6 μm mass-to-light ratios, $\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bul} = 0.7$, keeping points with $\delta V/V < 0.10$: 144 galaxies, 2681 points, of which 2352 lie beyond 2 kpc. The uncertainty on Δ combines the velocity error with a 0.1 dex stellar-population spread.

Against the *widest* kernel bound $C = 1.000$, 99.23% of points beyond 2 kpc comply at both footings. The eighteen $> 3\sigma$ exceptions lie in five galaxies: ESO 563-G021, NGC 5907, NGC 5985, NGC 6674 and UGC 12506. Thirteen are in NGC 5985 alone, which has $Q = 1$ and $i = 60°$ and so passes every standard cut; its $V_{\rm flat} = 294\,{\rm km\,s^{-1}}$ against a baryonic Tully–Fisher expectation near $195\,{\rm km\,s^{-1}}$ makes it a genuine exception, and we keep and name it rather than cutting it. Three of the remaining four are near edge-on ($i \geq 83°$) and contribute one or two points each.

The ceiling is tight enough to discriminate between kernels, and it does not favour the kernel this action carries. Table 2 compares the binned median excess with the exponential carrier's saturated ceiling and with $\nu_{\rm RAR}$.

$g_{\rm bar}/a_0$	measured Δ	carrier ceiling	$\nu_{\rm RAR}$	measured/carrier
1.29	0.816	0.368	0.610	2.2
1.70	0.994	0.368	0.633	2.7
2.34	0.979	0.368	0.647	2.7
3.27	1.190	0.368	0.641	3.2
4.44	0.750	0.368	0.615	2.0

Table 2: The transition region, where the ceiling is most exposed. Medians of Δ in logarithmic bins of $g_{\rm bar}$, beyond 2 kpc.

Beyond 2 kpc, 9.8% of points (263 of 2352, spread over 32 of 144 galaxies) exceed the exponential carrier's own ceiling a_0/e by more than 3σ. That is the ordinary population, not a tail. Two escapes are closed. A coherent shift of the stellar normalisation is pinned independently by the Newtonian limit, which every kernel shares: at $g_{\rm bar} > 10\,a_0$, where all kernels require $\Delta \to 0$, the measured median of $g_{\rm obs}/g_{\rm bar}$ over 90 points is 1.018, so the normalisation is already correct to 1.8%. Raising a_0 does not help either: the ceiling is a_0/e in physical units, and accommodating the largest measured excess would require $a_0 \geq 1.14 \times 10^{-9}\,{\rm m\,s^{-2}}$, twelve times the canonical value and far outside the 20% spread between the two footings. We note that a joint profiling of a_0 and Υ is the formal comparison and is not repeated here; what this statistic adds is that the coherent part of the Υ freedom is externally constrained.

5 Test on clusters

We use the twelve X-COP clusters, taking hydrostatic masses from the FORW reconstruction, gas masses from the published $f_{\rm gas}$ profiles, and stellar profiles for the seven clusters that ship one; the remaining five receive a radius-dependent stellar import measured from those seven and are excluded from every headline.

A radius-unit error had to be corrected first, and we verified it directly from the FITS headers rather than from any script: the gas profile's `RADIUS` column carries `TUNIT1 = R/R500`, with R_{500} given in each file's own header (for example 1250 kpc for A644), whereas earlier analysis multiplied it by 10^3 as if it were Mpc. Since R_{500} ranges from 1054 to 1430 kpc across the sample, that misplaced every gas measurement by a different per-cluster factor.

With the radii corrected, the median excess over the seven clusters with measured stars is

$$\Delta/a_0 = 3.37,\ 2.94,\ 2.48,\ 2.28,\ 2.10,\ 1.91,\ 1.63,\ 1.52,\ 0.99,\ 0.69$$

at $r = 40, 50, 75, 100, 150, 200, 300, 420, 750, 1000\,\mathrm{kpc}$. At 40 kpc that is 9.1 times the exponential carrier's exact bound a_0/e and 3.4 times the widest bound of any kernel in Table 1. *No choice of interpolation function can absorb this*, which is the point of proving the bound before measuring.

Both escapes close quantitatively. Restoring the ceiling by nonthermal pressure alone, with thermal and nonthermal components of similar logarithmic slope and $kT = 5\,\mathrm{keV}$, requires a one-dimensional turbulent velocity of $1155\,\mathrm{km\,s^{-1}}$ at 40 kpc falling to $673\,\mathrm{km\,s^{-1}}$ at 300 kpc, against the Hitomi and XRISM measurement of $164 \pm 10\,\mathrm{km\,s^{-1}}$ in the Perseus core: a factor 4.1 in velocity and 17 in pressure at the most favourable radius. Restoring it with unseen baryons requires 3.9 to 5.2 times the observed baryons inside 300 kpc, where the X-ray gas is directly imaged and dominant.

Clusters therefore require a genuine extra source. Because the bound is a theorem, this conclusion does not depend on a fit, on a choice of interpolation function, or on a dark-matter model.

6 What the cluster residual must be, and a null

Applying the kernel to the total source, the required source mass runs $M_{\rm src}/M_{\rm b} = 6.5$ at 40 kpc, 8.2 at 100 kpc and 5.2 at 750 kpc, a logarithmic slope of -0.05: it tracks the baryons. That looks like a strong constraint until the control is run. The published NFW fits to the same clusters give $M_{\rm NFW}/M_{\rm b}$ with slope $+0.03$, and the Newtonian hydrostatic ratio $M_{\rm HSE}/M_{\rm b}$ likewise $+0.03$, both equally flat. Baryon-tracing therefore does *not* discriminate between an extra source and a dark-matter halo, and we report it as a null. A rescaled acceleration scale does not fit either: the a_0 that would reproduce each cluster point runs from 24 to 8 times the canonical value with slope -0.32, so it is not a constant.

What survives as the discriminating statement is the ceiling itself, not the profile shape.

7 What is not claimed

The theorem is exact; everything else here is a measurement with stated systematics. Hydrostatic equilibrium is assumed for the cluster masses and is tested rather than granted in the nonthermal-support calculation, but the FORW reconstruction, the distance scale and the sphericity of these systems are inherited. The SPARC test uses fixed mass-to-light ratios; a per-galaxy stellar population model would move individual points, though the Newtonian-limit calibration bounds the coherent part of that freedom. This work does not derive a_0, does not derive $\kappa = \frac{1}{2}$, does not present a completed relativistic theory, and does not claim that any data set favours this framework over ΛCDM. Its two positive results are that the ceiling is a genuine parameter-free prediction of the modified-gravity class that a halo model cannot reproduce without fine-tuning, and that cluster cores violate it by a margin no interpolation function can absorb.

8 Reproducibility

Every number above is produced by a single committed script, `g03u_bounded_boost_theorem.py`, in the closure programme of the de Sitter–MOND repository, which carries nine checks that can fail and reports zero failures at the stated thresholds. The symbolic part of the theorem is verified in SymPy; the kernel suprema are computed on a fixed grid; the SPARC and X-COP inputs are the public data products, read with their own headers.